Experimental and Computational Study of Winglet Design Effects on Aerodynamic Stability, Lift, and Drag of a Missile

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The objective of this work (a paragraph) : the abstract will be written by the end of the semester, it is a summary of the project including the CFD results and the validation.

Nomenclature

A	=	amplitude of oscillation
a	=	cylinder diameter
C_p	=	pressure coefficient
Cx	=	force coefficient in the x direction
Cy	=	force coefficient in the y direction
c	=	chord
dt	=	time step
Fx	=	X component of the resultant pressure force acting on the vehicle
Fy	=	Y component of the resultant pressure force acting on the vehicle
f, g	=	generic functions
h	=	height
i	=	time index during navigation
j	=	waypoint index
K	=	trailing-edge (TE) nondimensional angular deflection rate

I. Introduction

CFD is a numerical computational-based method that is used to study and analyze the flow of fluids. CFD is widely used in aerospace and mechanical engineering as it builds on the foundation work for many projects that focus heavily on flow characteristics such as lift, drag, pressure, and general flow behavior around or inside systems. This allows for the reduction in cost and time as physical experiments can become expensive, especially at an industry level scale. This results in CFD playing an important role in modern engineering design by reducing cost, improving efficiency, and reducing the time taken up by traditional design processes, while enabling engineers to test over a range of different operation conditions. Although CFD is widely used as an industry standard tool, its results of simulated flow conditions are not always

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perfect nor correct because simulated ≠ real life conditions; therefore, it must be analyzed and interpreted carefully. Sources of error could come from multiple numerical and modeling choices within CFD, including mesh, time step, turbulence models, and boundary conditions. The CFD predictions acquired are heavily dependent on the setup, and even the slightest inaccuracies could lead to significant errors. At an industry level, that could have major consequences resulting in a loss of hundreds of thousands to millions of dollars. Therefore, it is essential to carefully analyze CFD outputs and validate them against experimental data to ensure reliability and engineering accuracy.[1][2]

Winglets are aerodynamic devices that can improve the performance and efficiency of high-speed vehicles by reducing drag and controlling airflow around the tips of wings or fins. In aerospace engineering, winglet designs are commonly studied using CFD to evaluate how different geometric parameters affect aerodynamic performance. For rockets and missiles, optimizing winglet geometry is important because these vehicles operate at high speeds where drag forces and shock waves significantly impact stability, fuel efficiency, and overall flight performance. Research on winglet configurations can help engineers design more efficient winglets and fin systems for rockets and missiles operating at high speeds [3]. Recent studies have also explored how winglet geometry can reduce sonic boom intensity while maintaining low drag characteristics at supersonic speeds. Parameters such as cant angle, sweepback angle, toe angle, and twist angle affect how shock waves and airflow interact with the vehicle during flight. These findings are especially relevant to rocket and missile applications because reducing drag and controlling shockwave behavior can improve aerodynamic stability and performance. By analyzing different winglet configurations through CFD simulations, engineers can develop designs that balance aerodynamic efficiency with reduced drag for high-speed flight vehicles.[4]

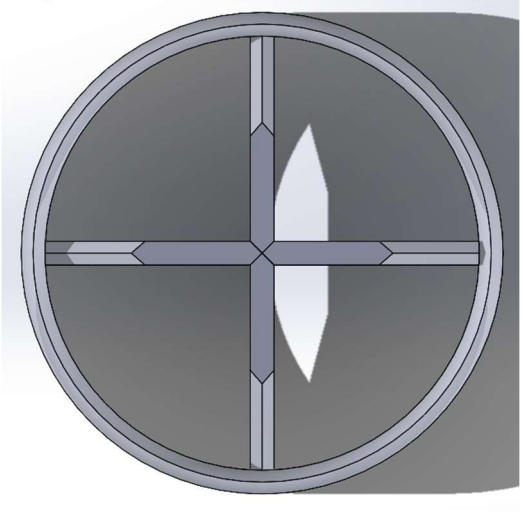
Winglets seem to be a good change to test out for conventional missiles. They are known to add lift to planes and have been studied extensively in the 1970s. However, they were not much needed then, but now they could be used to further increase the performance of missiles. Winglets add lift to planes when added to wings, which translates to increased efficiency and range. It could allow missiles to fly further or use less fuel for the same distance. With the increased lift, there could be a larger payload, which could further improve the efficiency and effectiveness of the missiles. Winglets also decrease the induced drag from wings, or in the case of this project, fins. This could also increase the top speed of the missile without modifications to the engine or anything else about the missile [5]. CFD simulations have been done to test improved cruise missile wings by changing multiple factors and have shown to greatly increase the lift-to-drag ratio. The tests, however, changed multiple different aspects of the wings, and cruise missile wings are a lot larger than a typical missile, if they even have wings at all. Winglets, however, were an addition to the improved wings that had a noticeable effect in the simulations. Winglets also tend to increase stability, which could be useful to make the missiles more accurate. [6]

Winglets affect aerodynamic loads, pressure distributions, structural bending, and overall wing performance. NASA tested how winglets affect all these factors using a low-speed wind tunnel, a transonic wind tunnel, and numerous scale models, including swept wings, winglets, ailerons, and nacelles. Their findings show that winglet loads are very similar to wing loads in sideslip. Furthermore, they have a large impact on the load distribution, specifically the aerodynamic forces acting at the wingtip. There was an increased load near the tip, largely due to the differing center of pressure compared to a normal wing. Ultimately, winglets improve aerodynamic efficiency at the price of increasingly complex loading, including bending or torques [7]. The cant angle is one of the most important design factors that must be considered for winglets, because they directly affect wing aerodynamics. They can change the lift and drag produced by the wing, as well as change the vortices that form at the wingtips. Computational fluid dynamics systems are ideal for visualizing these variables, and aeroacoustics simulations can provide insight into the noise production of winglets at various cant angles. Wings without winglets produce large vortices at the wingtip, winglets weaken these vortices which in turn reduce drag. The lift to drag ratio can

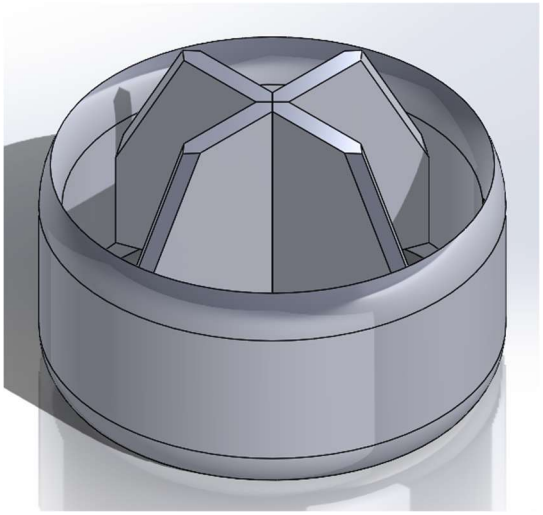
also be optimized by changing the cant angle, and it is likely that moderate cant angles are the ideal compromise for aerodynamic efficiency. [8]

II. Solid works model

Description: The fins have a large ring going around all four fins with a slightly larger diameter than the rocket. The ring has a smooth but sharp curve on the top and bottom for increased aerodynamics.

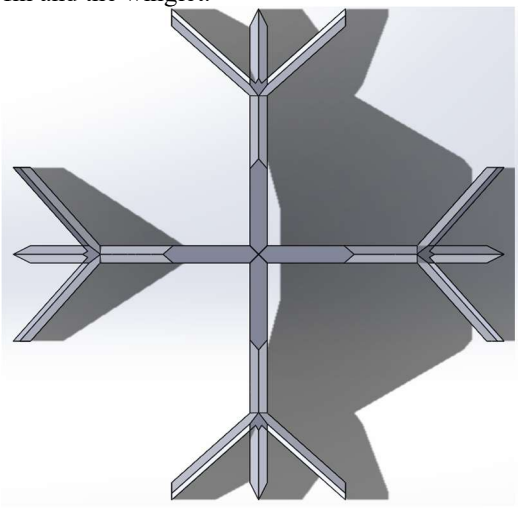


Top View

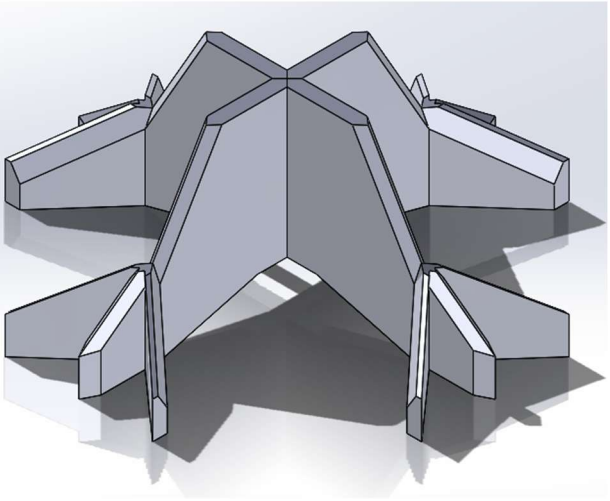


Isometric View

Description: The fins have two winglets extending diagonally from either face forming a 45 degree angle between the fin and the winglet.

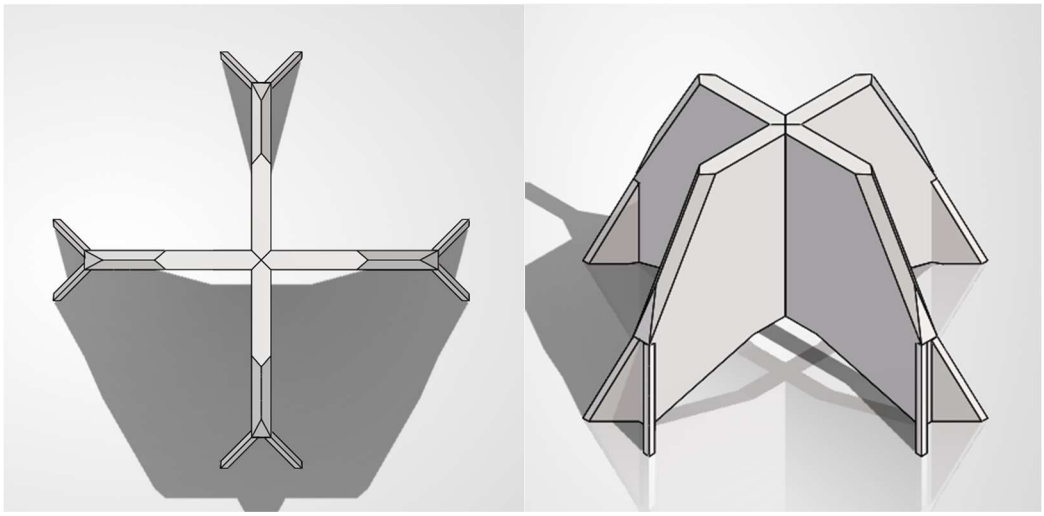


Top View



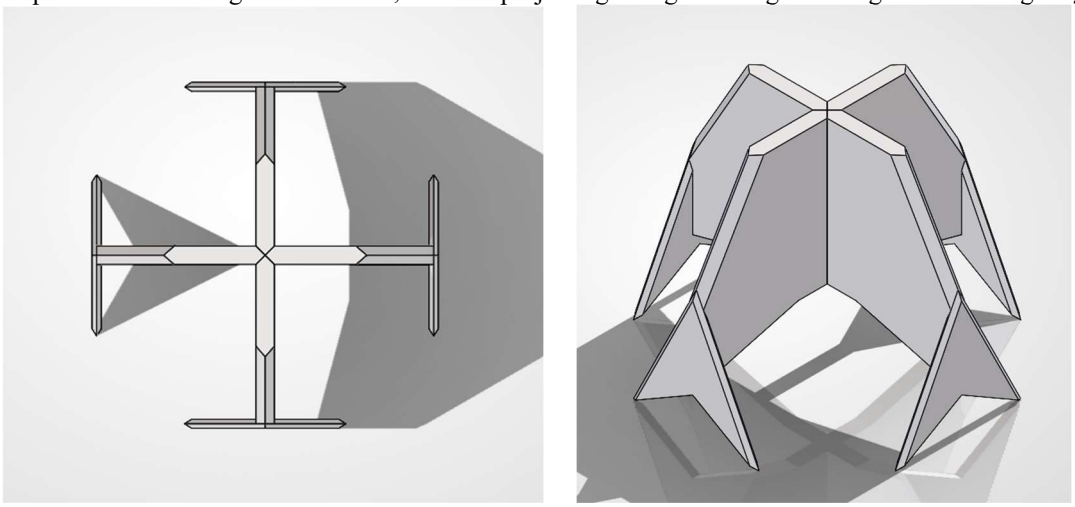
Isometric View

Description: The fin edges incorporate angled wingtip extensions perpendicular to each other, attached to the outer ends of each fin.



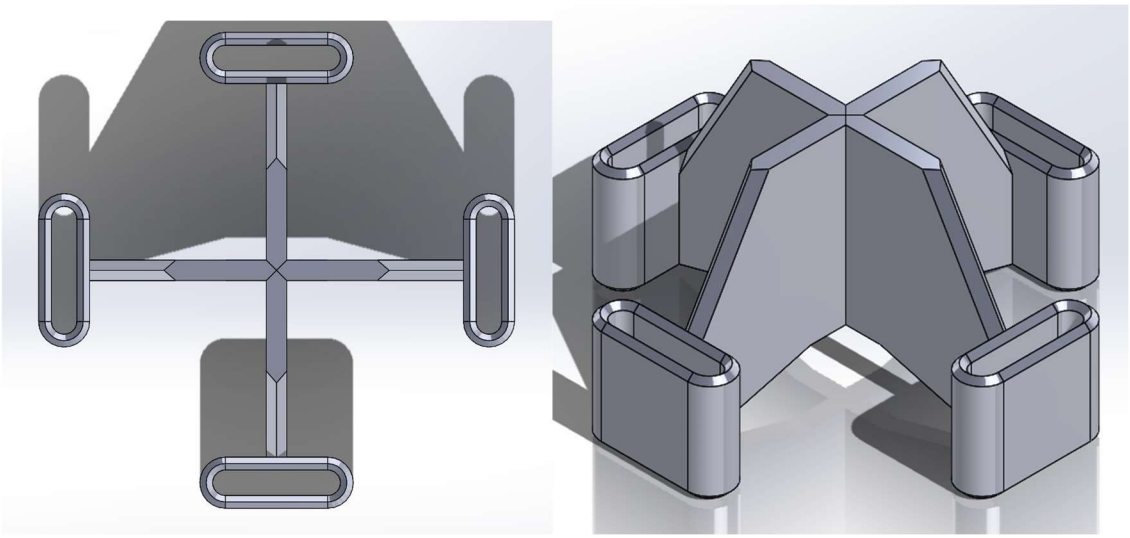
Top View Isometric View

Description: The fin design includes flat, outward projecting triangular winglets along the lower edge regions.



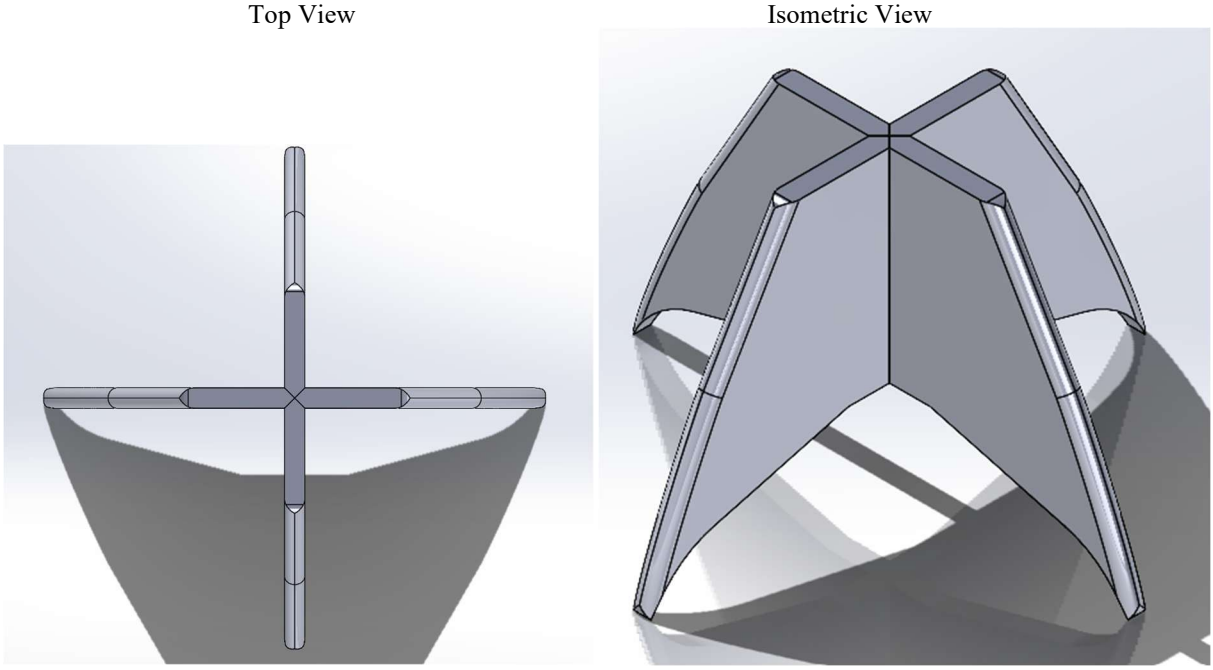
Top View Isometric View

Description: This fin design includes winglets that are slot shaped that goes along the lower fin area.

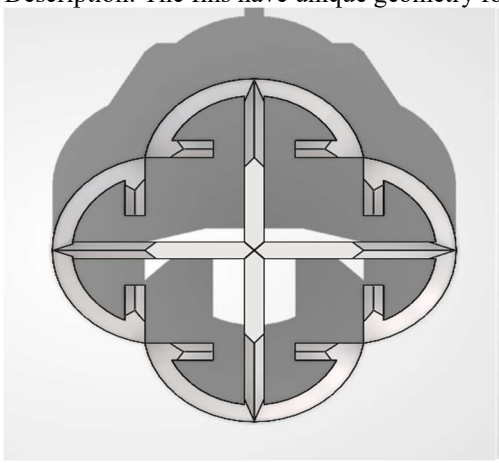


Description: This fin design includes winglets that are a simple raked design that continues from the base model.

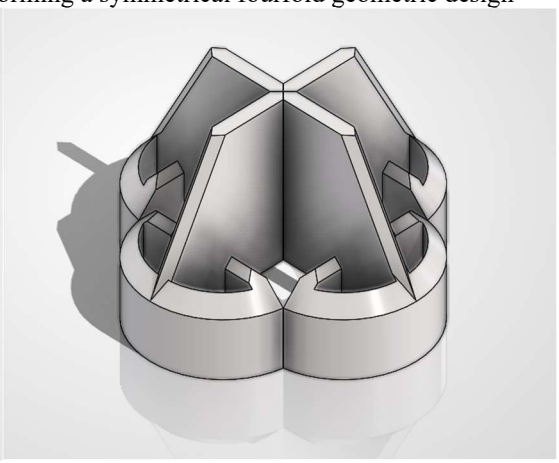
Top View Isometric View



Description: The fins have unique geometry forming a symmetrical fourfold geometric design

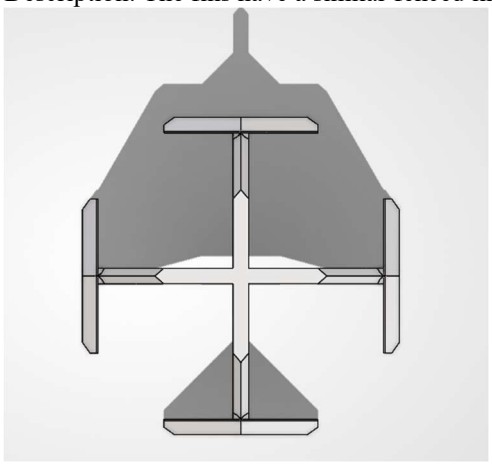


Top View

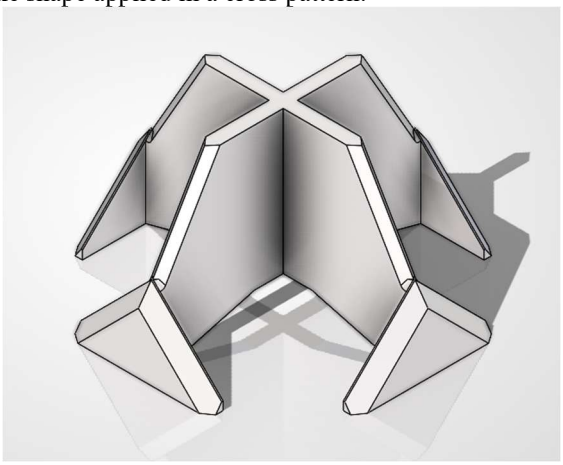


Isometric View

Description: The fins have a similar fenced like shape applied in a cross pattern.



Top View



Isometric View

III. CFD_analysis.

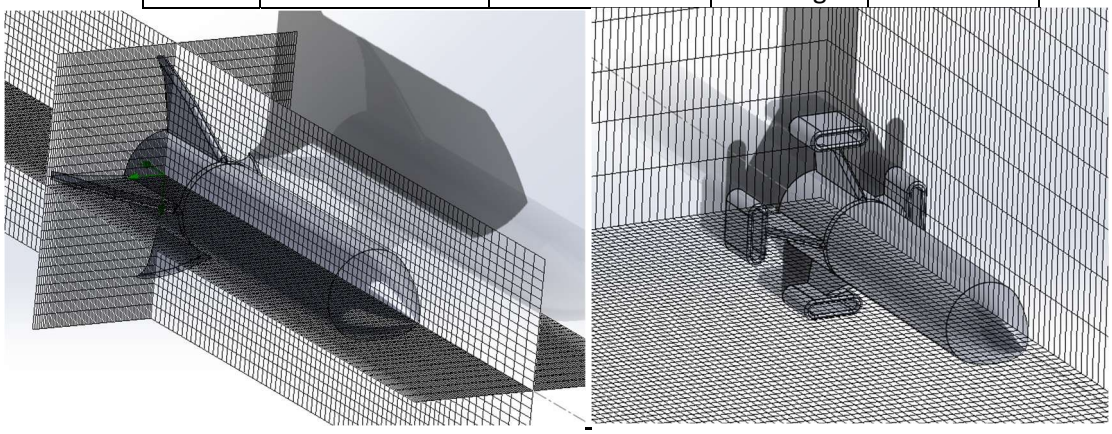
A. Boundary conditions

Create a table to show the boundary conditions

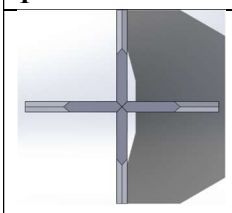
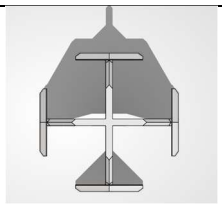
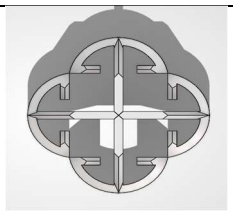
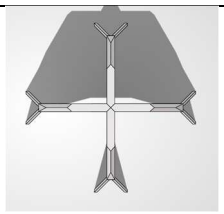
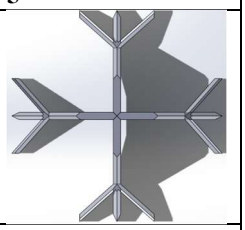
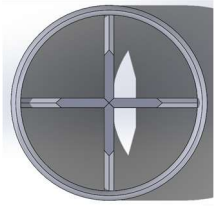
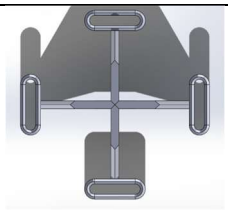
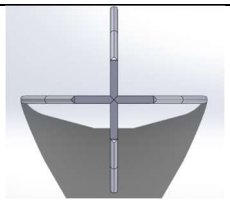
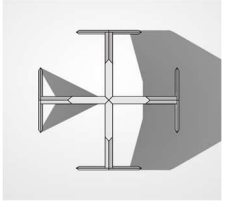
Boundary/Parameter Name	**Value/Condition**
Analysis Type	External Flow
Fluid	Air
Velocity	30 m/s
Ambient Pressure	101325 Pa
Ambient Temperature	293.2 K
Wall Conditions	Adiabatic Wall

B. Meshing

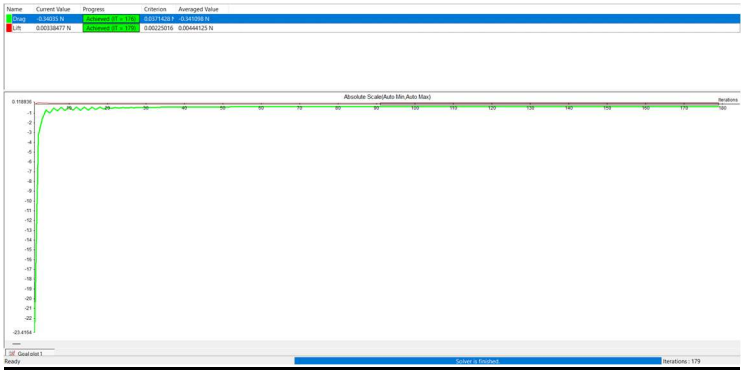
MESHING				
MODEL	*TOTAL # of CELLS*	*# of FLUID CELLS*	*CELL SHAPE*	*ITERATIONS*
1	676470	676470	Rectangle	179
2	756840	756840	Rectangle	188
3	1735687	1735687	Rectangle	338
4	710247	710247	Rectangle	182
5	760327	760327	Rectangle	183
6	1006151	1006151	Rectangle	364
7	830259	830259	Rectangle	229
8	643523	643523	Rectangle	242
9	712237	712237	Rectangle	185



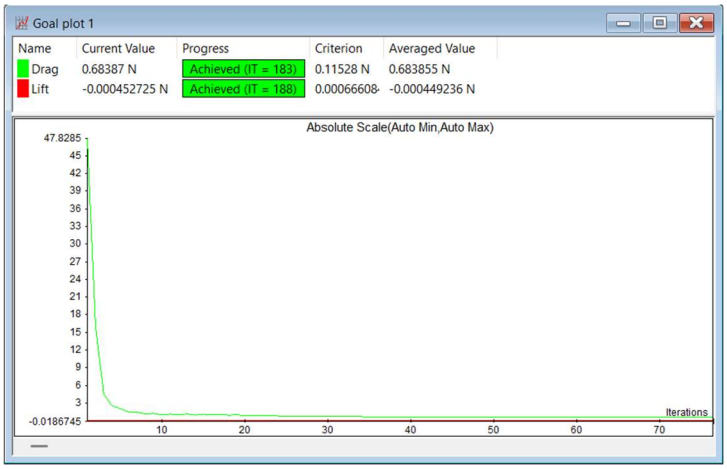
Goal Plots:

1	2	3	4	5
				
6	**7**	**8**	**9**	
				

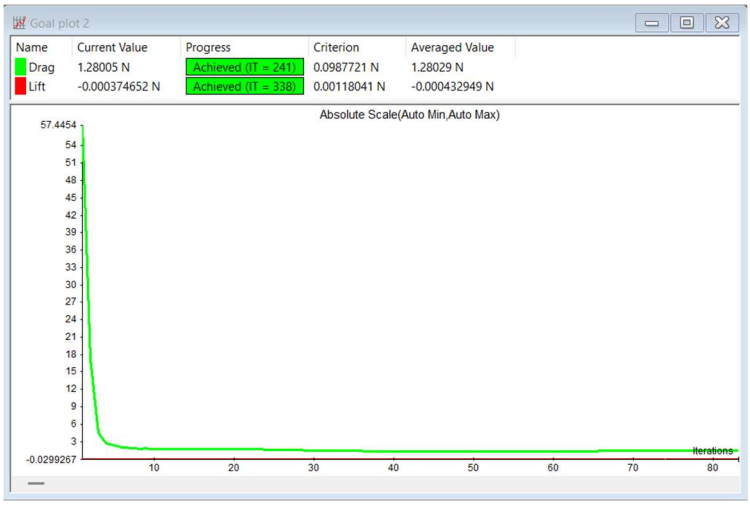
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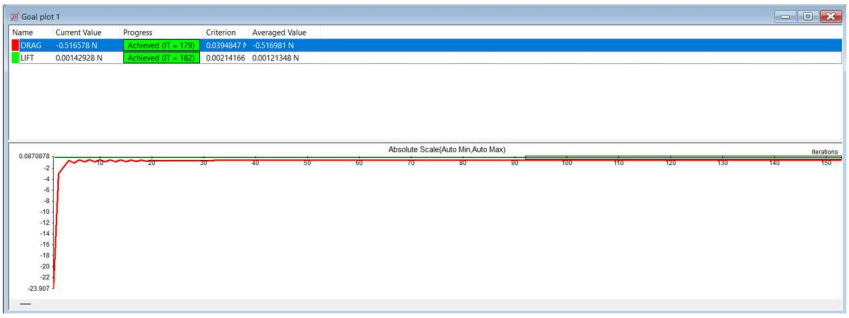
2.


Goal plot 1
Name
Current Value
Progress
Criterion
Averaged Value
Drag
0.68387 N
Achieved (IT = 183)
0.11528 N
0.683855 N
Lift
-0.000452725 N
Achieved (IT = 188)
0.00066608
-0.000449236 N
Absolute Scale(Auto Min,Auto Max)
47.8285
-0.0186745
Iterations
10
20
30
40
50
60
70

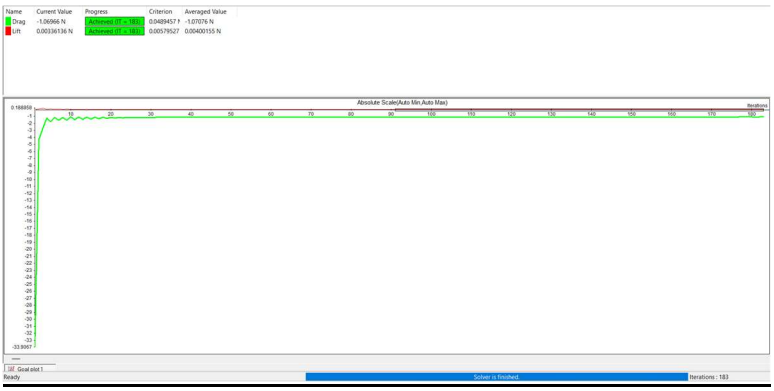
3.


Goal plot 2
Name
Current Value
Progress
Criterion
Averaged Value
Drag
1.28005 N
Achieved (IT = 241)
0.0987721 N
1.28029 N
Lift
-0.000374652 N
Achieved (IT = 338)
0.00118041 N
-0.000432949 N
Absolute Scale(Auto Min,Auto Max)
57.4454
-0.0299267
Iterations
10
20
30
40
50
60
70
80

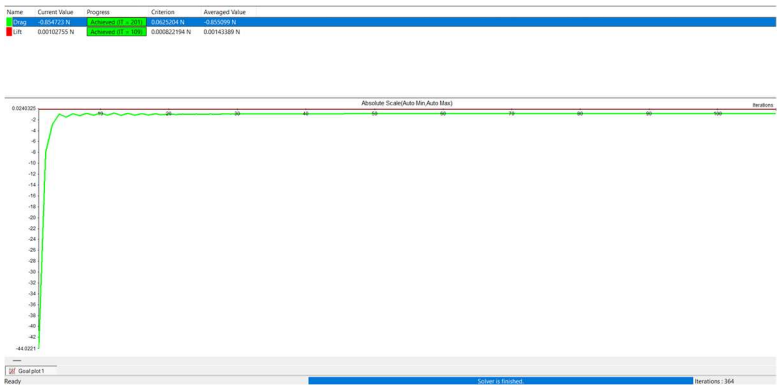
4.


Goal plot 1
Name
Current Value
Progress
Criterion
Averaged Value
DRAG
-0.516578 N
Achieved (IT = 170)
0.0394847 ?
-0.516981 N
LIFT
0.00142928 N
Achieved (IT = 182)
0.00214166
0.00121348 N
Absolute Scale(Auto Min,Auto Max)
Iterations

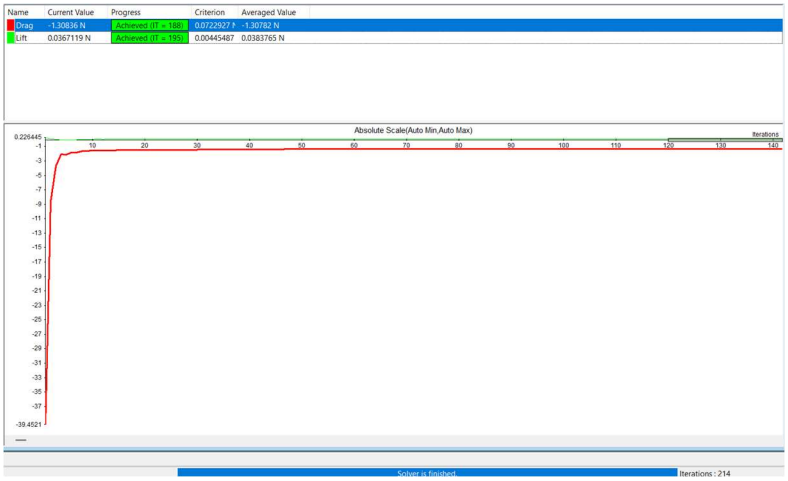
5.


Name
Current Value
Progress
Criterion
Averaged Value
Drag
-1.06966 N
Achieved (IT = 181)
0.0489457 ?
-1.07076 N
Lift
0.00336136 N
Achieved (IT = 181)
0.00579527
0.00400155 N
Absolute Scale(Auto Min,Auto Max)
Iterations
Goal plot 1
Ready
Solver is finished
Iterations : 183

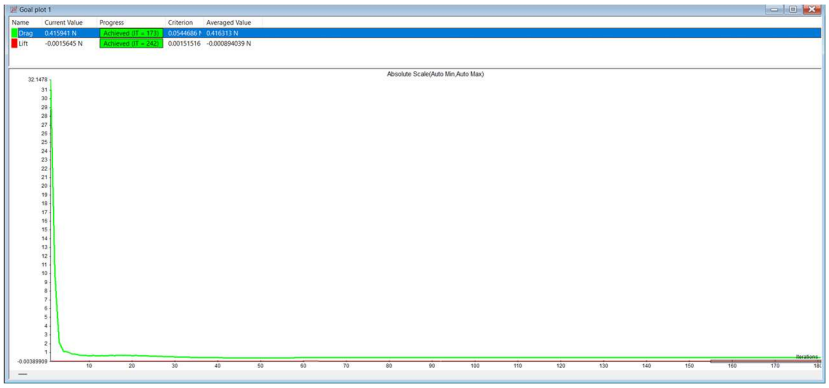
6.


Name
Current Value
Progress
Criterion
Averaged Value
Drag
-0.0414721 N
Achieved (IT = 331)
0.0052574 N
-0.0414399 N
Lift
0.00102755 N
Achieved (IT = 181)
0.000822194 N
0.00143389 N
Absolute Scale(Auto Min,Auto Max)
Iterations
Goal plot 1
Ready
Solver is finished
Iterations : 364

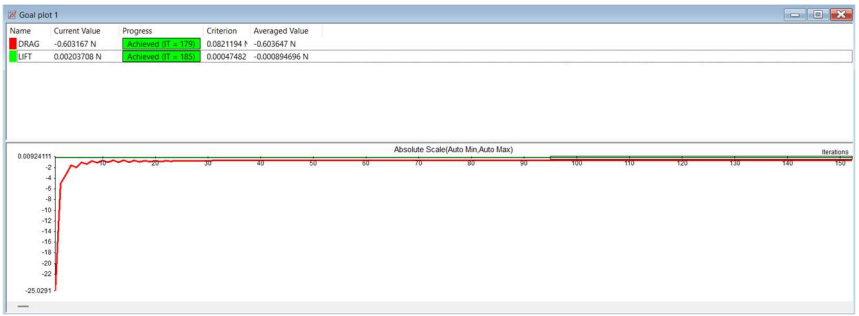
7.


Name Current Value Progress Criterion Averaged Value
Drag -1.30836 N Achieved (IT = 188) 0.0722927 N -1.30782 N
Lift 0.0367119 N Achieved (IT = 195) 0.00445487 0.0383765 N
Absolute Scale(Auto Min,Auto Max)
Iterations
Solver is finished.
Iterations : 214

8.

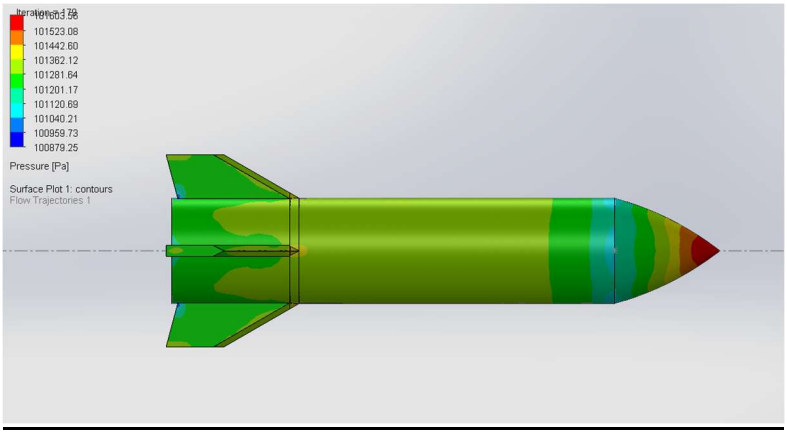

Goal plot 1
Name Current Value Progress Criterion Averaged Value
Drag 0.415941 N Achieved (IT = 171) 0.0544686 N 0.416313 N
Lift -0.0015645 N Achieved (IT = 247) 0.00151516 -0.000894039 N
Absolute Scale(Auto Min,Auto Max)
Iterations

9.

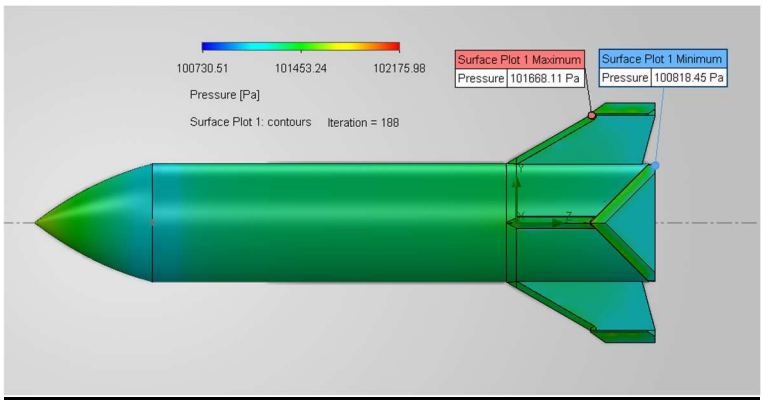

Goal plot 1
Name Current Value Progress Criterion Averaged Value
DRAG -0.603167 N Achieved (IT = 179) 0.0821194 N -0.603647 N
LIFT 0.00203708 N Achieved (IT = 185) 0.00047482 -0.000894696 N
Absolute Scale(Auto Min,Auto Max)
Iterations

Surface Plots:

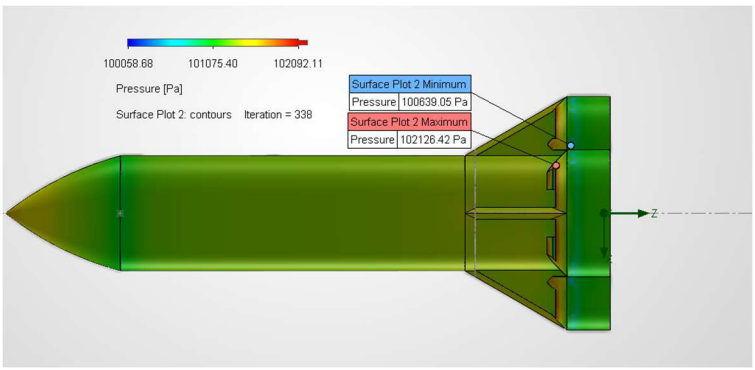
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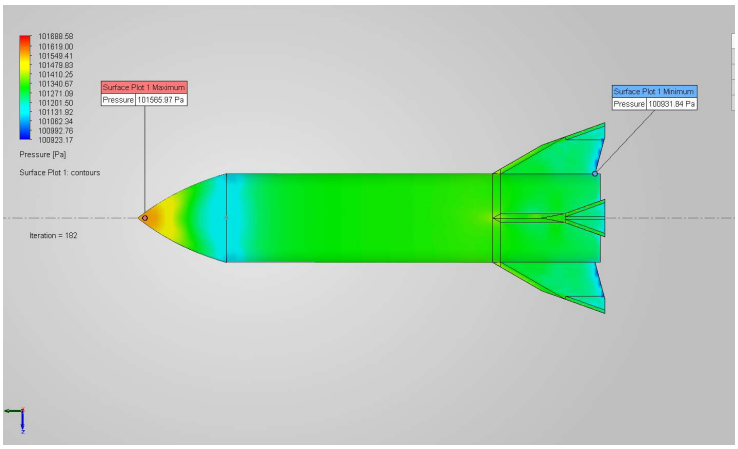
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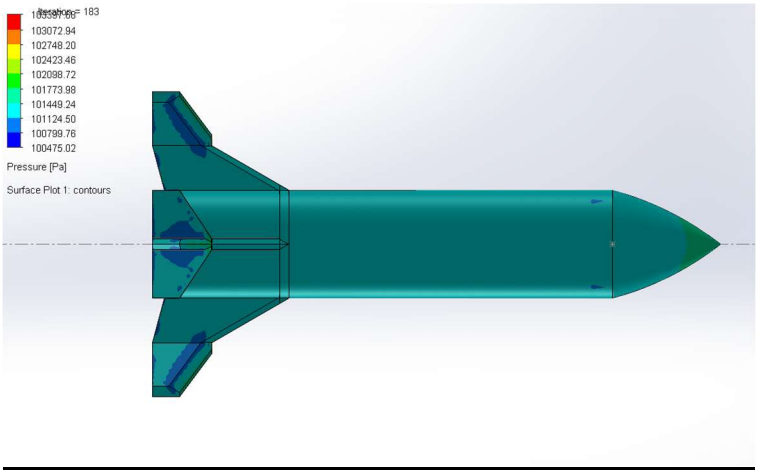
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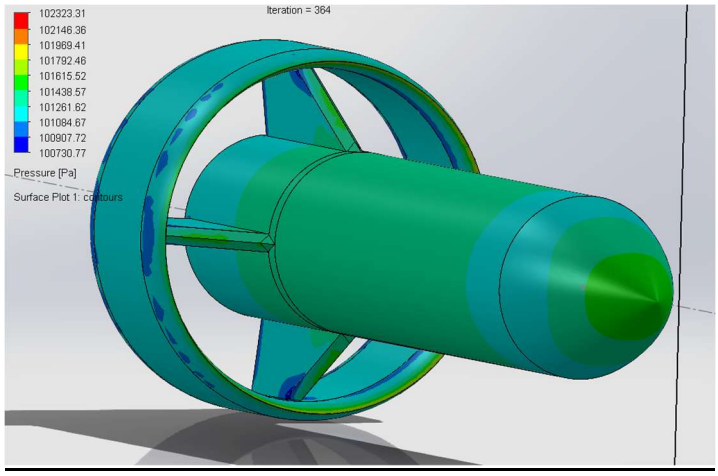
4.


Surface Plot 1 Maximum
Pressure 101565.97 Pa
Surface Plot 1 Minimum
Pressure 100931.84 Pa
Pressure [Pa]
Surface Plot 1: contours
Iteration = 182

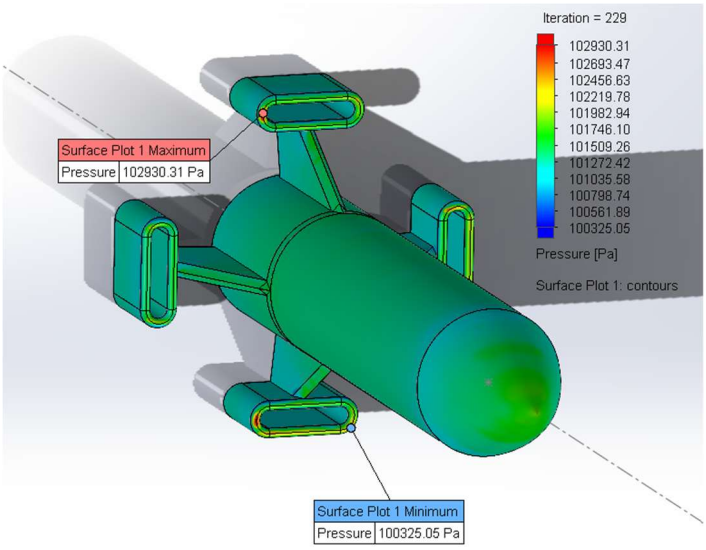
5.


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100799.76
100475.02
Pressure [Pa]
Surface Plot 1: contours

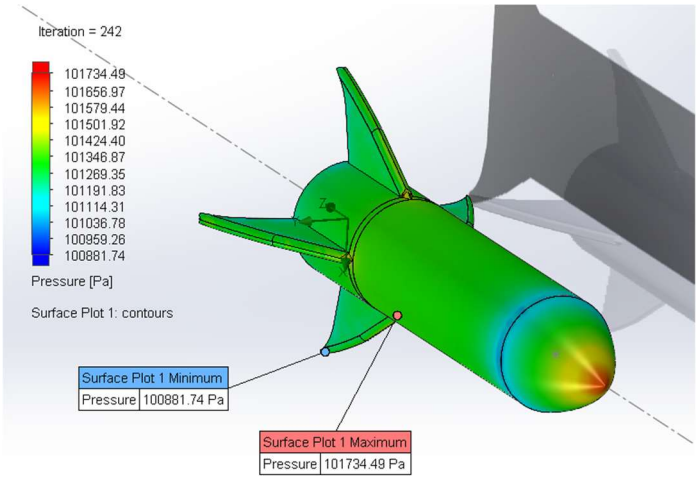
6.


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Pressure [Pa]
Surface Plot 1: contours

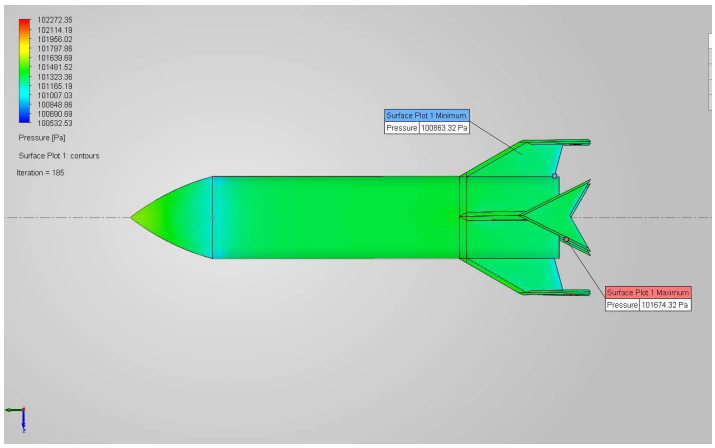
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Pressure [Pa]
Surface Plot 1: contours
Surface Plot 1 Maximum
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Surface Plot 1 Minimum
Pressure 100325.05 Pa

8.

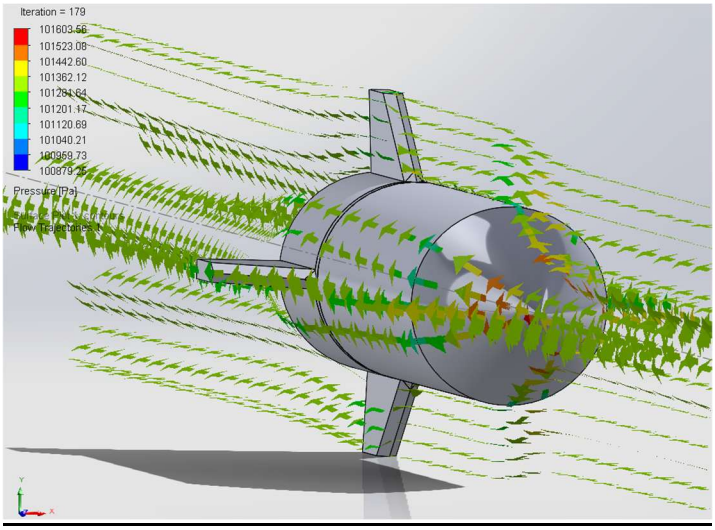

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101114.31
101036.78
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Pressure [Pa]
Surface Plot 1: contours
Surface Plot 1 Minimum
Pressure 100881.74 Pa
Surface Plot 1 Maximum
Pressure 101734.49 Pa

9.

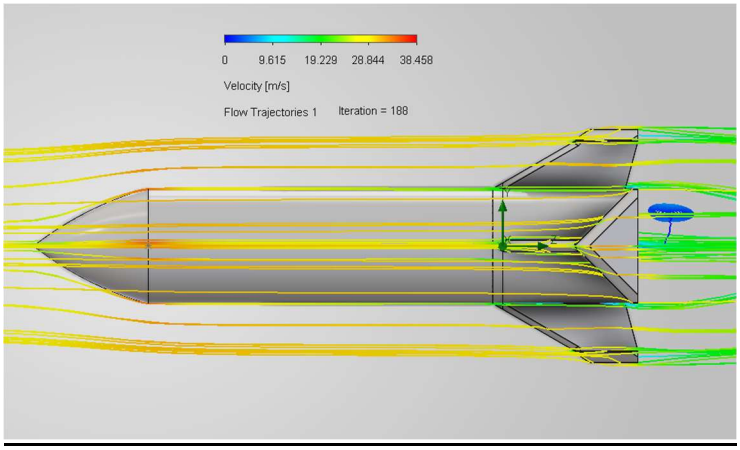

Pressure [Pa]
Surface Plot 1: contours
Surface Plot 1 Minimum
Surface Plot 1 Maximum

Flow Trajectories:

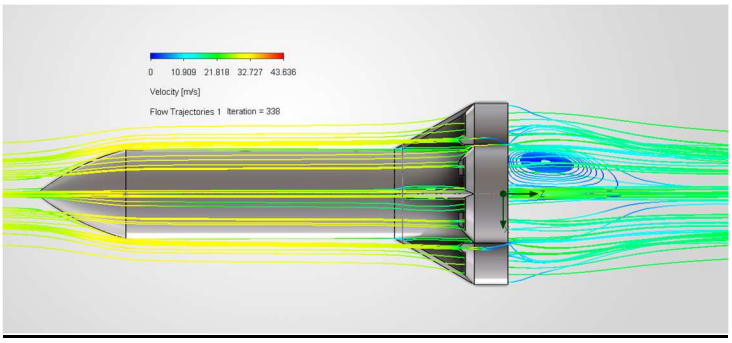
1.



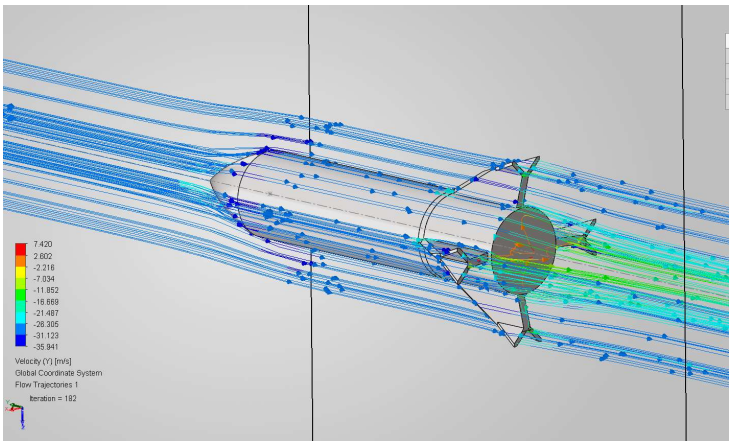
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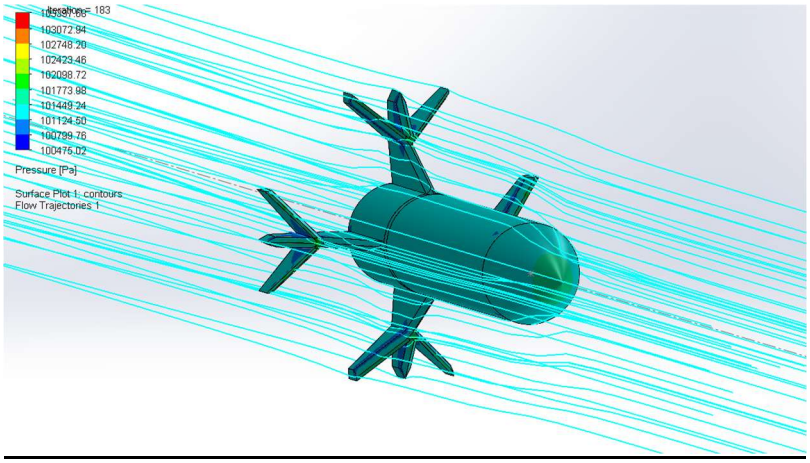
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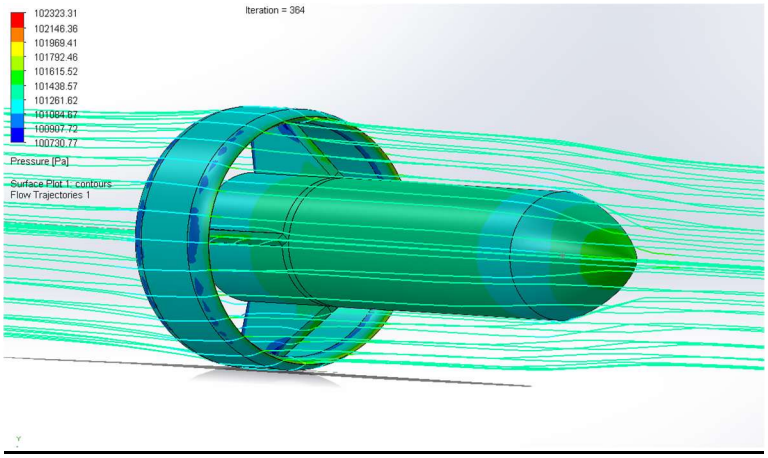
4.


7.420
2.602
-2.216
-7.034
-11.852
-16.669
-21.487
-26.305
-31.123
-35.941
Velocity (Y) [m/s]
Global Coordinate System
Flow Trajectories 1
Iteration = 182

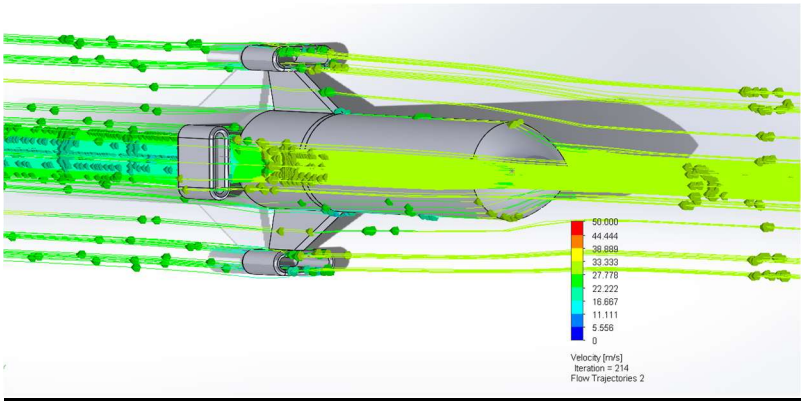
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102423.46
102098.72
101773.98
101449.24
101124.50
100799.76
100476.02
Pressure [Pa]
Surface Plot 1, contours
Flow Trajectories 1

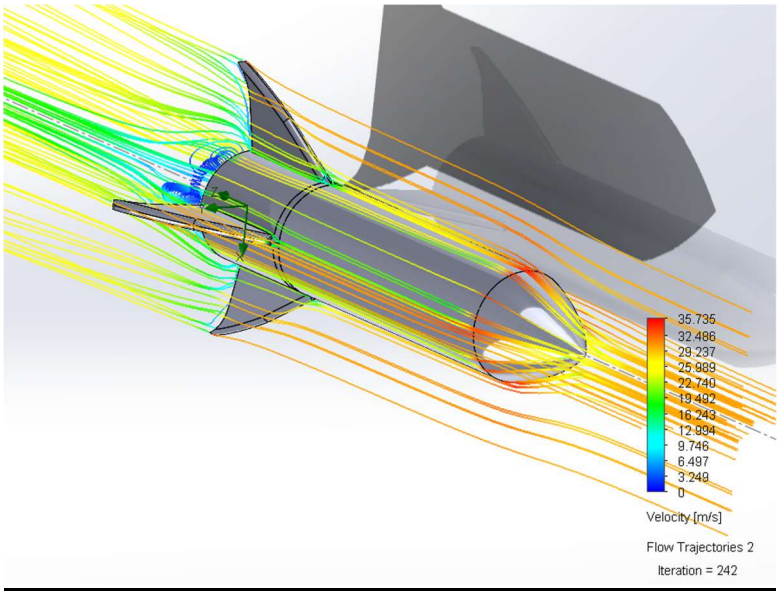
6.


Iteration = 364
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101261.62
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100730.77
Pressure [Pa]
Surface Plot 1, contours
Flow Trajectories 1

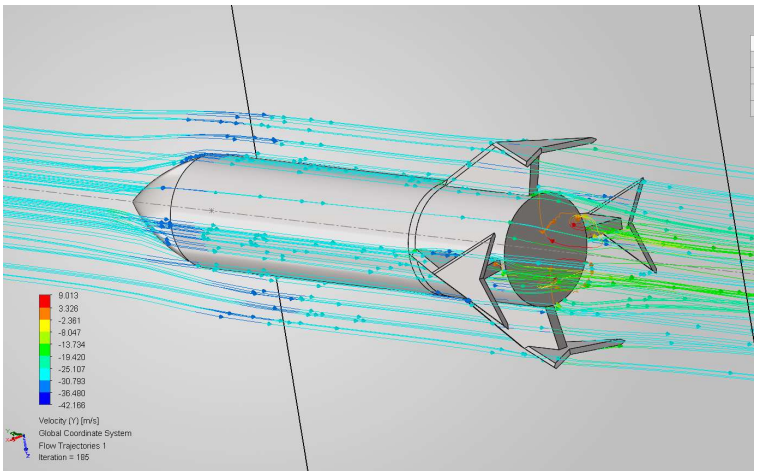
7.


50.000
44.444
38.889
33.333
27.778
22.222
16.667
11.111
5.556
0
Velocity [m/s]
Iteration = 214
Flow Trajectories 2

8.


35.735
32.486
29.237
25.989
22.740
19.492
16.243
12.994
9.746
6.497
3.249
0
Velocity [m/s]
Flow Trajectories 2
Iteration = 242

9.


9.013
3.326
-2.361
-8.047
-13.734
-19.420
-25.107
-30.793
-36.480
-42.166
Velocity (Y) [m/s]
Global Coordinate System
Flow Trajectories 1
Iteration = 185

C. Analysis

The convergence of the CFD results can be shown in each model's goal plot. In the plots, lift and drag values initially changed as the flow developed around each model, as the flow developed the solution gradually stabilizes and as the number of iterations continued to increase. The aerodynamic forces for lift and drag, despite lift being almost negligible at an angle of attack of 0 degrees, the values level out and reach a near constant value. If iteration continued, we would get minimal changes in our lift and drag solutions which means the CFD had reached convergence. The flow contours shown in the figures provide insight into the behavior of the flow being generated by each unique geometry. The velocity flow contours show acceleration and deceleration of airflow around the surfaces of each model. The pressure surface plots show loading caused by the interaction of incoming air flow and wing geometry creating higher points of pressure on leading edges and lower points of pressure at regions where flow accelerated over the surface at 0 degrees. Overall, the converged results, velocity and pressure plots provided insight into aerodynamic effect of symmetrical winglet configurations at an attack angle of 0 degrees.

D. Discussion

After running all the models through simulations, the results show that there is almost no lift created. This would make sense as the winglets are all symmetrical so any lift created would be canceled out by downforce. The slight lift values that were observed during the simulations could be due to the air flow not being perfect, especially in more turbulent environments. The drag values varied a bit more between the different models. The highest drag was recorded on model 7 with the lowest drag from model 1. Many of the models also created vortexes behind the fins which was unexpected, however the winglets also seemed to slow down the air around the winglets frequently.

IV. Building Prototype and Testing

Eight model rockets were 3D-printed at the Wentworth Makerspace, each with the same body but different winglet geometries, along with a ninth base rocket with no winglets. They were all printed in black PLA utilizing a 5% infill to decrease weight. In order to secure each rocket to the supporting rod in the wind tunnel, a through-hole was drilled through the center of mass of each rocket. Then a soldering iron was used to push a small threaded insert into the drilled hole, on the surface of the rocket body. This method ensured that it was not possible for the rocket to detach from the supporting rod. With this method, the supporting rod passed through the entire body of the rocket, through the threaded insert, and was secured with a small nut of negligible frontal area. Although this placed the rocket slightly below the center of the wind tunnel, this complicated fixture process was necessary, as 5% infill does not provide sufficient support to reliably retain the insert. In order to measure the drag and lift forces of each design, the rocket was placed in the wind tunnel, and the airflow velocity was slowly ramped up to 30 mph. The peak drag and lift forces were recorded, corrected by subtracting the drag and lift contributions of the supporting rod. Any imperfections in the data can be attributed to small variations in the angle of attack because the angle of attack changes the frontal area of the rockets, as well as affecting the rocket's ability to generate lift.







Sample Calculations:

Lift Coefficient:

$$C_L = \frac{L}{\frac{1}{2}\rho V^2 A_p} \quad (1)$$

$$0.0025 = \frac{0.0034}{\frac{1}{2} * 1.22 * 30^2 * 0.00245}$$

Drag Coefficient:

$$C_D = \frac{D}{\frac{1}{2}\rho V^2 A_f} \quad (2)$$

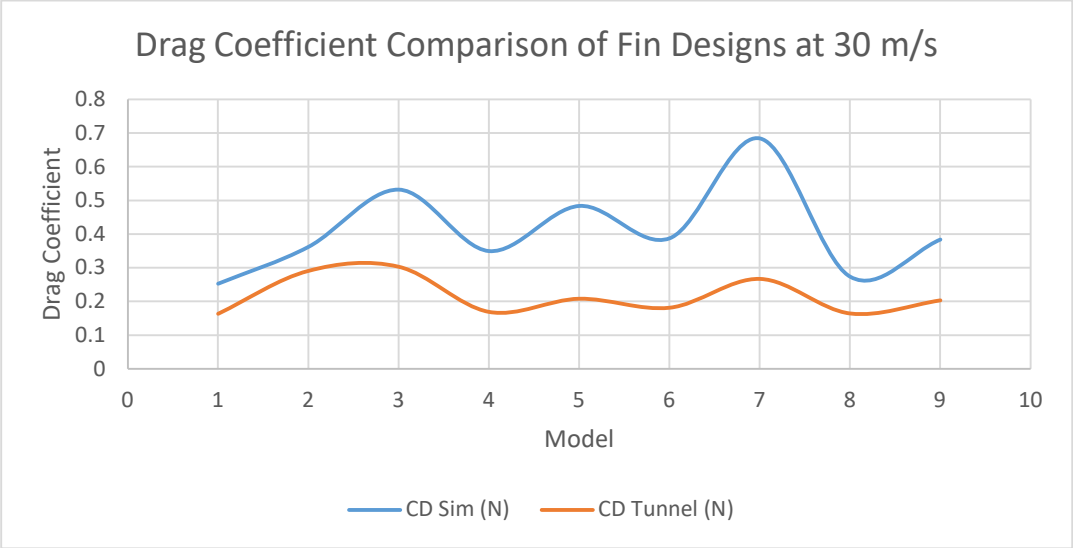
$$0.253 = \frac{0.34}{\frac{1}{2} * 1.22 * 30^2 * 0.00245}$$

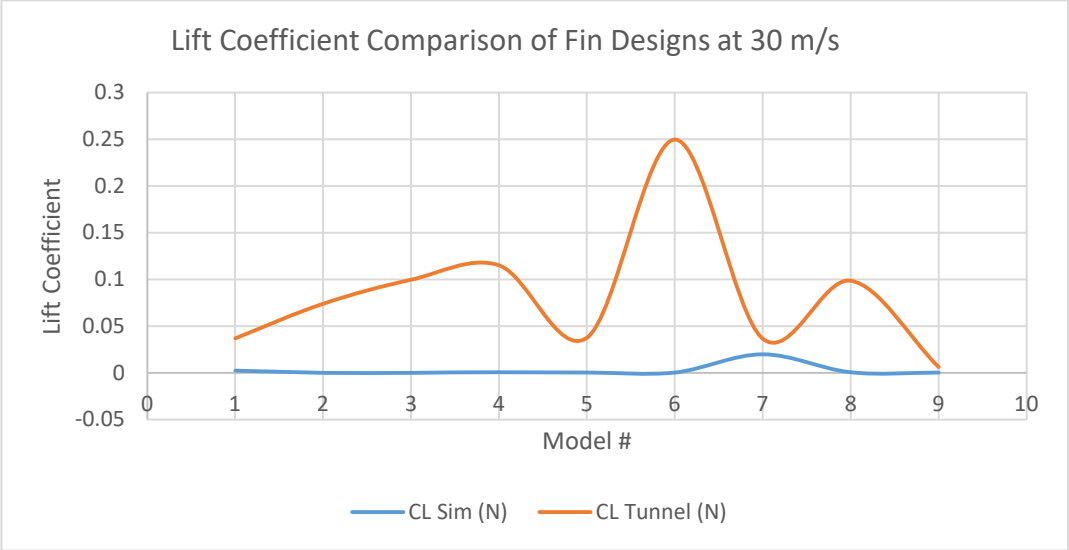
Summarized Results:

Simulation			
MODEL	*VELOCITY*	*DRAG (N)*	*LIFT (N)*
1	30 m/s	0.34035	0.00338477
2	30 m/s	0.683855	0.000449236
3	30 m/s	1.28029	0.000432949
4	30 m/s	0.516981	0.00121348
5	30 m/s	1.02313	0.00280933
6	30 m/s	0.854723	0.00102755
7	30 m/s	1.30782	0.0383765
8	30 m/s	0.416313	0.000894
9	30 m/s	0.603647	0.000894696

EXPERIMENTAL WIND TUNNEL (Corrected)			
MODEL	*VELOCITY*	*DRAG (N)*	*LIFT (N)*
1	30 m/s	0.22	-0.05
2	30 m/s	0.55	0.14
3	30 m/s	0.73	0.24
4	30 m/s	0.25	0.17
5	30 m/s	0.44	-0.08
6	30 m/s	0.4	0.55
7	30 m/s	0.51	-0.07
8	30 m/s	0.25	0.15
9	30 m/s	0.32	0.01

V. Validation





Comparing the results of the experiment to the simulations was rather difficult due to the suboptimal experimental conditions. The results were very off compared to the simulations, with the lowest percent error being at 24% but the rest were over double that error rate. There could be many reasons as to why the errors were so great. One of the reasons why could be the fact that the experiments were rather rushed; we only had enough time to run the experiment once per model, leaving a small sample size. This drastically reduced the quality of the data because any outliers would be the only result. Another issue with the experiment was the fact that we had to consider the drag and lift of the rod that was holding the model in the wind tunnel, when doing this we assumed the rod was fully exposed to the air of the tunnel which was not the case. During our experiments the rod was going through the model therefore exposing much less surface area to the wind causing us to overcorrect for it in the calculations. Another issue is that the process for getting the rods attached to the model was very troublesome; the model was printed with very thin walls, causing the heat insert to not hold onto the model well. To fix this, we had to drill a hole through the entire model and slide the rod through and occasionally use a nut on the other end to hold it in place. This alone caused more inaccuracies compared to the simulation, but there was also the issue that the model was free to rotate. Due to the way that it was mounted the rod was free to spin slightly in each direction, thus when the wind tunnel starts the model sometimes turns more to one side, changing the amount of drag or lift by changing the exposed surface area. The

models also were not perfectly flat, due to the difficulty in mounting the models some of the models were at a slight angle also changing the exposed surface area and thus the lift and drag. Besides all of the errors that could have potentially been avoided with more time and equipment there are some that are unable to be controlled. For example, the wind tunnel itself has walls that somewhat affect the flow of the air changing the results a bit. Another difference could be the surface roughness of the model, because they were 3d printed, it is likely that they had some small bumps and ridges at the layer lines that would affect the airflow.

Fortunately, despite the many limitations of the experimental process, there was some agreement between the wind tunnel measurements and the simulation results. Most notably, both the experimental and simulated data indicated that the lift generated by each rocket design was minimal and essentially negligible. This outcome was expected because every model had a perfectly symmetrical cross section. As a result, the airflow over the top and bottom surfaces of the fins and winglets traveled under nearly identical conditions, producing equal pressure distributions on both sides. These balanced pressure forces effectively canceled one another, resulting in virtually zero net lift.

However, the results related to drag did not support the expected benefits of winglets. In theory, winglets reduce drag by weakening wingtip vortices and decreasing the associated induced drag. Contrary to this expectation, both the experimental data and the simulations showed that the rocket without winglets produced the lowest coefficient of drag. Several factors could explain this unexpected result, with the most likely being the relatively low air velocity used in both the computational fluid dynamics simulations and the wind tunnel testing. At lower airspeeds, wingtip vortices are generally less intense, reducing the amount of induced drag they create. Consequently, the aerodynamic advantage provided by winglets is also diminished, making their effect on overall drag difficult to observe under the testing conditions used in this study.

Lift and Drag Coefficients:

Drag Coefficient			
Model	*CD Sim (N)*	*CD Tunnel (N)*	*% Error*
1	0.2530389	0.163562693	54.70455
2	0.3621039	0.291227178	24.33727
3	0.5324292	0.303582271	75.38219
4	0.350066	0.169283794	106.7924
5	0.4840584	0.2081707	132.5295
6	0.3882475	0.181695125	113.6808
7	0.6845361	0.266943031	156.4353
8	0.2737587	0.164394731	66.5252
9	0.3844543	0.203803482	88.63969

Lift Coefficient			
Model	*CL Sim (N)*	*CL Tunnel (N)*	*% Error*
1	0.0025165	0.037173339	93.23046
2	0.0002379	0.074130554	99.67912
3	0.00018	0.09980787	99.8196
4	0.0008217	0.11511298	99.28619
5	0.0004861	0.037849218	98.71556
6	0.0004668	0.249830796	99.81317
7	0.0200869	0.03663924	45.17643
8	0.0005879	0.098636839	99.404
9	0.0005698	0.006368859	91.05304

VI. Conclusion

The computational and experimental methods performed provided insight into the aerodynamic effects of winglets on missile fins. Despite our CFD model not producing the same magnitude of lift and drag coefficients measured during the wind tunnel testing, it successfully identified the performance trends between different winglet configurations. This demonstrates the value of computational simulation as a design and screening tool before physical testing.

The results emphasize the importance of experimental validation when evaluating aerodynamic performance, as real-world flow conditions have effects that are difficult to capture with simplified numerical models. The results showed that model 1, which did not have any winglets, produced the lowest drag of all the designs tested. Although model 8 performed the best among the designs with the winglets (likely due to its similarity in geometry to model 1), its drag was still higher than the original base design. This indicated that, under the conditions tested, adding winglets did not improve the overall aerodynamic performance of the missile. Since the model is designed to travel primarily in the forward direction and does not require the additional lift that winglets are commonly used to provide on aircraft, the added winglet surfaces were not beneficial in this application. Instead, the additional geometry introduced by the winglets increased drag without providing a significant aerodynamic advantage. The nearly negligible lift measured in both the simulations and the wind tunnel testing also supports this conclusion, as the symmetrical rocket design did not require additional lift to achieve the intended behavior.

The methodology used in this study emphasizes the importance of experimental validation when evaluating aerodynamic performance, as real-world flow conditions have different effects that are difficult to capture with simplified computational testing. The wind tunnel experiment showed that the actual performance of the design differed significantly from the CFD predictions, demonstrating that simulations do not always perfectly represent real-life conditions. Factors such as the supporting rod, slight changes in angle of attack, the ability of the models to rotate during testing, and the surface imperfections from 3D orienting contributed to the differences in the testing results. However, both methods showed similar overall behavior, particularly the minimal lift generated by the symmetrical designs. Lack of time was also a factor contributing to the experimental results, as only one test was run per model in the wind tunnel. Having a minimum of three tests per model would allow for more reliable results. The methods learned from this experiment can be applied to the design and optimization of aerodynamic surfaces. Additionally, this methodology can be used for other similar studies. The future work for this study should investigate additional winglet geometries, improved turbulence models, and a wider range of operating conditions to further enhance prediction accuracy and aerodynamic performance. Ultimately, the experiment concluded that the addition of winglets to rocket and missile designs does not reduce drag and does not increase aerodynamic efficiency.

Acknowledgments

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